

STANFORD AI FOR LEAN4 CLUB

Inverting Path Compression

a proof-digestion case study

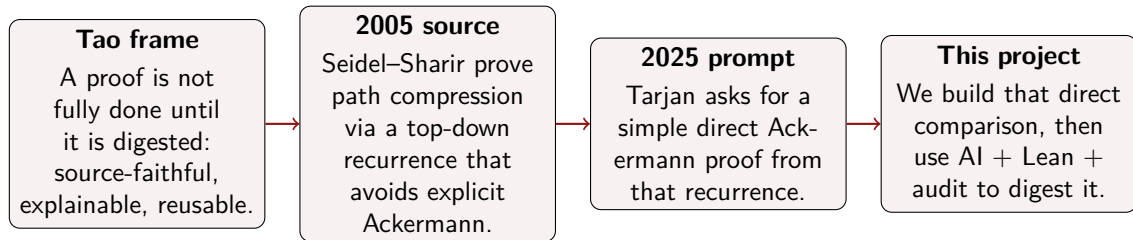
Invert the Seidel–Sharir shrinkers; the recursion becomes Ackermann growth.

Don Poindexter

After Seidel–Sharir (2005), Tarjan's Dagstuhl prompt (2025), and a targeted prior-art audit

TL;DR: the whole story

A proof-digestion loop around a concrete mathematical prompt



High-level punchline

Invert the rank-shrinkers. In the inverted coordinates, Seidel–Sharir's recursive shrinker becomes Ackermann growth.

Where the problem comes from

Seidel–Sharir supplied the recurrence; Tarjan asked for the direct Ackermann bridge

2005: Seidel–Sharir

Top-Down Analysis of Path Compression,
SIAM J. Comput. 34(3), 515–525.

They give a top-down recurrence based on a hierarchy J_k ; their abstract says the inverse-Ackermann bound follows *without having to introduce the Ackermann function itself*.

Source: Seidel–Sharir, DOI 10.1137/S0097539703439088.

2025: Tarjan at Dagstuhl 25191

The Dagstuhl summary records that Tarjan drew attention to different analyses of path compression and raised the question of an alternative proof.

Section 5.3 points back to the Seidel–Sharir recurrence and asks for the direct classical-Ackermann connection.

Source: Dagstuhl Seminar 25191 report page and Section 5.3.

The task is to **connect Seidel–Sharir's J hierarchy directly to classical Ackermann growth.**

Why believe this is a live prompt?

The targeted novelty search found adjacent routes; this bridge was not located

Source family	What it gives	Gap relative to this route
Tarjan-style analyses	Classical Ackermann appears directly in a bottom-up analysis.	Uses classical A directly, rather than passing through the Seidel–Sharir $J^\diamond$ recurrence.
Seidel–Sharir / Seidel	A top-down J -hierarchy that makes the bound natural.	Keeps the story internal to the J -hierarchy, rather than proving a quantitative $J \rightarrow A$ bridge.
Modern simplifications and formalizations	Cleaner union-find proofs and verified analyses.	Works in other frameworks rather than the route $J_k \mapsto R_k \mapsto A$.

Cautious conclusion

The targeted audit found many adjacent analyses, but no public source with the specific threshold-inversion route used here.

What specific bridge was missing?

The object and inequality that make the direct comparison work

The new coordinate

For the Seidel–Sharir hierarchy, define the max-inverse

$$R_k(t) = \max\{r \geq 0 : J_k(r) \leq t\}.$$

This asks: at level k , how large a rank can be pushed below threshold t ?

The direct bridge to classical Ackermann

The proof note shows

$$R_{z+1}(Q) \geq A(z, 4Q) \quad (z \geq 1, Q \geq 1).$$

So the Seidel–Sharir shrinkers dominate classical Ackermann growth after inversion.

Evidence-based claim: public prior-art search found no source proving this bridge for $J^\diamond$.

Why this belongs in a Lean/AI club

Tao's proof-digestion frame: three jobs; the third is the least automatic

1
Generation

2
Verification

3
Digestion

Talk thesis

This talk is about using AI, Lean, and red-team audit to support digestion without outsourcing judgment. The math is the example; the meta-claim is the workflow.

“A theorem is not done until it is source-faithful, narratable, and reusable.”

Minimal background

Enough vocabulary to read the recurrence

Path compression

Union–find stores each set as a rooted tree. A find operation walks from a node to the root, then rewires the visited nodes closer to the root.

The analysis counts the total cost of many such compressions.

Ranks and the cost function

Ranks measure how high a node can sit in the forest. Let

$$f(m, n, r)$$

be the worst-case cost of m compressions on n nodes when the maximum rank is at most r .

What Seidel–Sharir add

They build a hierarchy $J_0, J_1, J_2, \dots$ of rank-shrinking functions. If $J_k(r)$ is small, the recurrence gives a good cost bound.

The setup

The Seidel–Sharir hierarchy, written explicitly

Start with a simple rank shrinker

$$J_0(r) = \left\lceil \frac{r-1}{2} \right\rceil.$$

The diamond operation

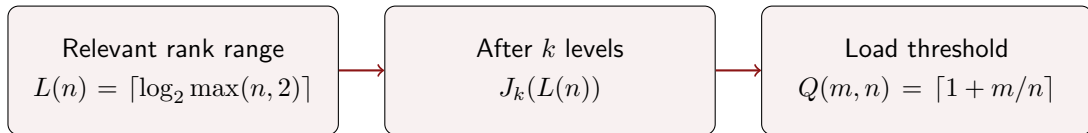
For any shrinker g , define a new shrinker $g^\diamond$ by

$$g^\diamond(r) = \begin{cases} g(r), & g(r) \leq 1, \\ 1 + g^\diamond(\lceil \log_2 g(r) \rceil), & g(r) > 1. \end{cases}$$

Then Seidel–Sharir set $J_{k+1} = J_k^\diamond$.

What the recurrence asks us to prove

The path-compression bound is reduced to understanding how fast J_k shrinks



Cost recurrence

$$f(m, n, r) \leq k m + 2n J_k(r).$$

So once $J_k(r)$ is small, the cost bound follows.

Core question

How small does k need to be before

$$J_k(L(n)) \leq Q(m, n)?$$

The trick: invert shrinking into reach

Comparing a shrinker to a grower directly fights the objects

Shrink view

Given rank r , how small does level k make it?

$$r \longmapsto J_k(r).$$

Reach view: the digestion move

Given threshold t , how far up does level k reach?

$$R_k(t) = \max\{r \geq 0 : J_k(r) \leq t\}.$$

Base row is exact:

$$R_0(t) = 2t + 1, \quad J_0(r) \leq t \iff r \leq 2t + 1.$$

The engine

The diamond recursion becomes row-by-row exponential amplification

On the recursive branch $J_k(r) > 1$:

$$J_{k+1}(r) = 1 + J_{k+1}(\lceil \log_2 J_k(r) \rceil).$$

To prove $J_{k+1}(r) \leq t + 1$, it suffices that

$$\lceil \log_2 J_k(r) \rceil \leq R_{k+1}(t),$$

which is forced by

$$J_k(r) \leq 2^{R_{k+1}(t)}.$$

Threshold step

$$R_{k+1}(t + 1) \geq R_k\left(2^{R_{k+1}(t)}\right).$$

One extra threshold unit at level $k + 1$ buys an exponentially larger input threshold at level k .

Why Ackermann appears

Row-feeding recurrence + Ackermann recursion: same structural shape

Row domination

$$R_j(x) \geq A(j+1, x) \quad (j \geq 1, x \geq 1).$$

Main comparison: the answer to Tarjan's prompt

$$R_{z+1}(Q) \geq A(z, 4Q) \quad (z \geq 1, Q \geq 1).$$

Where the $4Q$ comes from

For $Q \geq 2$:

$$\begin{aligned} R_{k+1}(Q) &\geq R_k\left(2^{R_{k+1}(Q-1)}\right), \\ R_{k+1}(Q-1) &\geq R_0(Q-1) = 2Q-1, \\ 2^{2Q-1} &\geq 4Q. \end{aligned}$$

The $Q = 1$ endpoint is handled separately in the proof note.

Closing the cost bound

Choose the Ackermann row that clears the rank range; plug into the source recurrence

$$L(n) = \lceil \log_2 \max(n, 2) \rceil, \quad Q(m, n) = \left\lceil 1 + \frac{m}{n} \right\rceil.$$
$$\alpha_Q(m, n) = \min\{z \geq 1 : A(z, 4Q(m, n)) > L(n)\}.$$

Main comparison gives

$$J_{z+1}(L(n)) \leq Q(m, n) \quad \text{for } z = \alpha_Q(m, n).$$

Therefore

$$f(m, n, L(n)) \leq (\alpha_Q(m, n) + 3)m + 4n = O(m \alpha_Q(m, n) + n).$$

For the real source threshold $1 + m/n$, the nonintegral case costs at most one extra level.

The proof spine

Everything exists to support the threshold step and the main comparison

Source recurrence $f \leq km + 2nJ_k(r)$

Source hierarchy $J_0 = \lceil (r-1)/2 \rceil$, $J_{k+1} = J_k^\diamond$

Basic J package: monotone, unbounded, $J_{k+1} \leq J_k$, R_k finite

Threshold step $R_{k+1}(t+1) \geq R_k(2^{R_{k+1}(t)})$



Threshold jump $R_{k+1}(Q) \geq R_k(4Q)$ for $Q \geq 2$

Row domination $R_j(x) \geq A(j+1, x)$ for $j, x \geq 1$

Main comparison $R_{z+1}(Q) \geq A(z, 4Q)$ for $z, Q \geq 1$



Alpha/cost consequence $f(m, n, L(n)) \leq (\alpha_Q + 3)m + 4n$

★ = the two slides everything else exists to support.

Lesson 1. Digestion is the bottleneck

AI generates artifacts cheaply. The bottleneck is the story that survives audit.

Raw workflow produced

- prompts
- proof notes ($v0.1.0 \rightarrow v0.1.4$)
- adversarial audits
- source-anchor packet
- finite sanity-check script
- Lean formalization branches
- Codex worker traces

What survives

Invert the Seidel–Sharir shrinkers;
diamond becomes Ackermann
growth.

The compression ratio matters: every hour of generation is wasted if no human can carry the result.

Lesson 2. Formalization exposes architecture

A “primitive” lemma turned out to be derived

Schematic before

```
structure ThresholdAssumptions where
  baseExact
  monotoneThreshold
  monotoneLevel
  thresholdStep
  thresholdJump -- primitive
```

Schematic after

```
structure ThresholdCoreAssumptions where
  baseExact
  monotoneThreshold
  monotoneLevel
  thresholdStep

theorem threshold_jump_from_step :
  -- derived from core assumptions
```

A “magic move” became a structural consequence. That is proof digestion in its strongest form.

Commit milestone: b29a6a9.

Lesson 3. Honest boundaries

What is machine-checked, and what is still a paper bridge?

Lean-checked: abstract engine

- packet Ackermann lemmas
- row-domination invariant under `ThresholdCoreAssumptions`
- threshold jump derived from threshold step + monotonicity + base exactness
- main comparison $A(z, 4Q) \leq R(z + 1, Q)$ for $z, Q \geq 1$

Paper-only: concrete bridge

- concrete J_0 and $J_{k+1} = J_k^\diamond$
- concrete inverse
 $R_k(t) = \max\{r : J_k(r) \leq t\}$
- proof that concrete R satisfies `ThresholdCoreAssumptions`
- α_Q , α_J^Q , α_J^S , and cost consequence

Strategic target: concrete $R \Rightarrow \text{ThresholdCoreAssumptions}$.

First steps: define concrete J_0 , diamond, and concrete R_k .

Two things to remember

Mathematical TL;DR

Invert the Seidel–Sharir shrinkers; diamond becomes Ackermann growth.

Proof-digestion TL;DR

Lean and adversarial audit helped separate the proof into a checked abstract engine and an explicit concrete bridge target.

github.com/patternscientist/path-compression-digestion

Questions?

Sources and reproducibility

- Raimund Seidel and Micha Sharir, *Top-Down Analysis of Path Compression*, SIAM J. Comput. 34(3):515–525, 2005. DOI: [10.1137/S0097539703439088](https://doi.org/10.1137/S0097539703439088).
- Robert E. Tarjan, “Top-down analysis of path compression,” in Dagstuhl Seminar 25191, *Adaptive and Scalable Data Structures*, Dagstuhl Reports 15(5), 2025.
- Targeted novelty memo: prior-art search over Seidel–Sharir, Seidel slides/lectures, Tarjan-style analyses, modern simplifications, verified union-find work, and Dagstuhl follow-up traces.
- Project repository: github.com/patternscientist/path-compression-digestion. Start with the proof note, audit, sanity script, and source-anchor packet.

Novelty hygiene

The deck states an evidence-based claim: a targeted public prior-art search found no source with this specific threshold-inversion bridge. It does not claim to rule out every private note, lecture, or unindexed thesis.